

HELLO, WORLD!
(LECTURE)



OUR FIRST PROGRAM

Take a look at our very first program below!

```
print("Hello World!")
```

We saw that this prints out "Hello World!" to the terminal.

SPECIAL CHARACTERS

Take a look at our very first program below!

```
print("Hello World!")
```

We saw that this prints out "Hello World!" to the terminal.

But, what if we wanted to print out another interesting quote?

```
"It takes a great much of pulling," said the dog-shot man.
```



SPECIAL CHARACTERS

Let's try it out:

```
print("""It takes a great much of pulling," said the dog-shot man."")
```

SPECIAL CHARACTERS

Let's try it out:

```
print("""It takes a great much of pulling," said the dog-shot man."")
```

So, that totally didn't work. What gives?

ESCAPING SPECIAL CHARACTERS

In Python, quotation marks `"` are used to mark the start and end of a string.

If we want to actually print a quotation mark, we need a special way to tell python that it is *a part of the string*, rather than indicating its end.

ESCAPING SPECIAL CHARACTERS

In Python, quotation marks `"` are used to mark the start and end of a string.

If we want to actually print a quotation mark, we need a special way to tell python that it is *a part of the string*, rather than indicating it's end.

We can do this by **escaping** the `"` with a `\`. As an example:

```
print("\\"It takes a great much of pulling,\" said the dog-shot man.")
```

`\` tells the computer to treat the next character as "special". In other words, it allows it to be a part of the string itself! Neat.

LECTURE ACTIVITIES

Which of these lines of code work?

M1 `print("his heart was his fate\\")`

- A: Works; B: Doesn't Work

M2 `print("howdy\\\"\\\" partner\\\"")`

- A: Works; B: Doesn't Work

S7 (🤔🤔🤔)

How can we print the `\` character itself? *Hint: `print("\\")` does not work.*

SPECIAL CHARACTERS: WHY

We did those activities with special characters to get you in the habit of running code, looking at the output, evaluating correctness/errors, and paying attention to small details. Those are the important skills; the actual mechanics of `\` are somewhat niche.

CALLING FUNCTIONS

To get Python to do something interesting, we will often **call a function**!

- Functions are special portions of code that accomplish something for us.
- Our first function in CIS1100 is `print()`
- We might call a function in order to...
 - Make something happen *to the computer* (e.g. printing to the terminal)
 - Calculate some result
 - or both!

A function call looks like: `function_name()` (Parenthesis at end $\leftrightarrow$ calling)

PRINT()

The `print()` function prints things out so that we can see them in the terminal. Within the `()` we must provide the string we want it to print.

```
# prints the string "Hello World!"  
print("Hello World!")
```

Reminder: `#` indicates the start of a comment in python

LEN()

The `len()` function calculates the **length** of the thing we pass in within the `()`.

For example, `len("Hi")` is `2` because `"Hi"` has two characters. In other words, its length is 2 characters!

On its own `len()` doesn't do much. So this program ends up just printing hello:

```
len("hello")    # doesn't print anything, just calculates the length
print("hello")
```

LEN() AND PRINT()

To make `len()` share its results, use it with `print()`:

```
print(len("Train Dreams")) # prints 12
```

`len("Train Dreams")` *evaluates* to the number `12`, then we pass that into `print()` to print it.

Note: `len(print("Train Dreams"))` is incorrect because `print` evaluates to nothing.

INPUT()

`input()` is a very useful function that does two things.

It both *prints* what you pass in and *evaluates* to what the user typed in.

Let's take a look at the following code:

```
print(input("What is your name?"))
```

First, this program prints out "What is your name".

Then, it reads what you type in (*after you hit enter*), and evaluates to that.

LECTURE ACTIVITIES

S8 What gets printed?

```
print(len("How are you?"))
```

S9 What gets printed?

```
print(len("Harry said, \"No shot.\"))
```

L11 Describe what the program below does.

```
print(len(input("Give me something: ")))
```

FUNCTIONS WITH MULTIPLE INPUTS

Some functions, like `print()`, can take in multiple inputs.

When we would like to pass in multiple inputs, we use commas (`,`) to separate them:

```
print("Hob in the Corner", "and the Boneless Man")
```



```
Hob in the Corner and the Boneless Man
```

When passing in more than one string, they will be printed together *with a space between them.*

EXPLAINING NAME = INPUT()

In the pre-lecture video, we did something like:

```
name = input("What is your name?")
```

Let's dissect this a little bit!

INTRO TO VARIABLES

Lets start with a simpler example:

```
name = "Joel"
```

This creates a **variable** called

name holding the value **"Joel"**

You can think of a variable like a box with a name attached to it. This box is named **name**

- the value in the box can change.
- the name of the box

name

Joel

PRINTING VARIABLES

We can print variables similar to how we print strings.

```
name = "Joel"  
print(name) # prints Joel
```

When we pass in the name of the variable, this tells python to see what is stored "inside that box".

name

Joel

VARIABLES CHANGE OVER TIME

Variables can change over the course of a program, and can only hold one value.

Consider:

```
name = "Harry"      <-----  
name = "Joel"  
print(name)
```

*Reminder: Code runs from top
to bottom!*

name

Harry

VARIABLES CHANGE OVER TIME

Variables can change over the course of a program, and can only hold one value.

Consider:

```
name = "Harry"  
name = "Joel"          <-----  
print(name)
```

*Reminder: Code runs from top
to bottom!*

name

Joel

EXPLAINING NAME = INPUT

Consider the following:

```
name = input("What is your name? ")  
print("Hello,", name, "!")
```

This will:

- ask you your name
- read what you typed in, and store it into a variable ("Box") called `name`
- prints out `"Hello, [name] !"` where `[name]` is what you typed in!

LECTURE ACTIVITY

C12 What does this program do?

```
first_name = input("First Name: ")
last_name = input("Last Name: ")

print("Hi! My name is", first_name, "last_name")
print("My first name is", len(first_name), "letters long.")
print(" ")
print("Good bye!")
```

REMINDER:

- No lecture on Monday, next lecture on Wednesday, September 3rd.
- There is another check-in due before that lecture as well.
- Office Hours will start next week (keep an eye on Ed)
- No recitation next week, those start on Sep 8 and Sep 9.